Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: UHI Team
b. Team leader name: Ahmad Syafiq Kamil
c. Problem Statement:
   "Urban heat data already exists, but it is not easily translated into location decisions, intervention choices, priorities for vulnerable residents, and accountable policy documents"

Brief about the idea

FROM HEAT MAPS TO POLICY ACTION

TEDUH is an AI agent that helps city governments identify the hottest and most vulnerable areas, assess potential interventions, and turn the findings into policy drafts.

Planners can ask questions in Indonesian/ English. **TEDUH** connects data, models, simulations, evidence, and policy briefs in one workflow

01 IDENTIFY

Land Surface Temperature (LST) maps and a Heat Vulnerability Index (HVI).

02 TEST

Simulate trees, cool roofs, and green roofs.

03 PRIORITIZE

Build intervention packages within budget and equity constraints.

04 DRAFT

A policy brief with traceable sources for every figure.

Residents: Roof Check feature home-cooling recommendations

Opportunities

The Gap TEDUH Addresses

DATA WITHOUT A PATH TO POLICY

Academic heat maps rarely inform detailed spatial plans, regional development plans, or local programs

TOOLS ARE TOO TECHNICAL

Simulation and optimization often require specialized skills and software

UNEQUAL EXPOSURE TO RISK

Low-income residents, older adults, young children, and outdoor workers need to be prioritized

TEDUH'S Unique Value

1. One conversation: heat data → simulation → budget portfolio → policy brief;
2. Explicit equity: prioritize the highest Heat Vulnerability Index (HVI) quintile; default allocation of 60% of the budget.
3. Traceable figures: deterministic tools + number-provenance validation.
4. Citizen participation: photo-based Roof Check with safety-constrained recommendations

List of features offered by the solution

Heat map & HVI

LST, anomalies, NDVI, building cover, population density, and vulnerable groups.

Sub-district profile

Rankings, HVI components, and key drivers (SHAP).

AI agent chat (ID / EN)

Ask in plain Indonesian or English; answers highlight areas on the map.

Scenario simulation

Trees, cool roofs, and green roofs; indicative ΔLST results with literature-based range.

Budget portfolio

Intervention packages aligned with budgets and equity constraints.

Policy brief

Structured draft and PDF with a source for every figure.

Data sources & model card

Attribution, cross-validation metrics and limitations on /sumber-data.

Roof check

Roof photo → Gemini Vision classification → rule-based recommendations

Privacy by design

Roof photos are never stored; EXIF is stripped; location kept at kelurahan level.

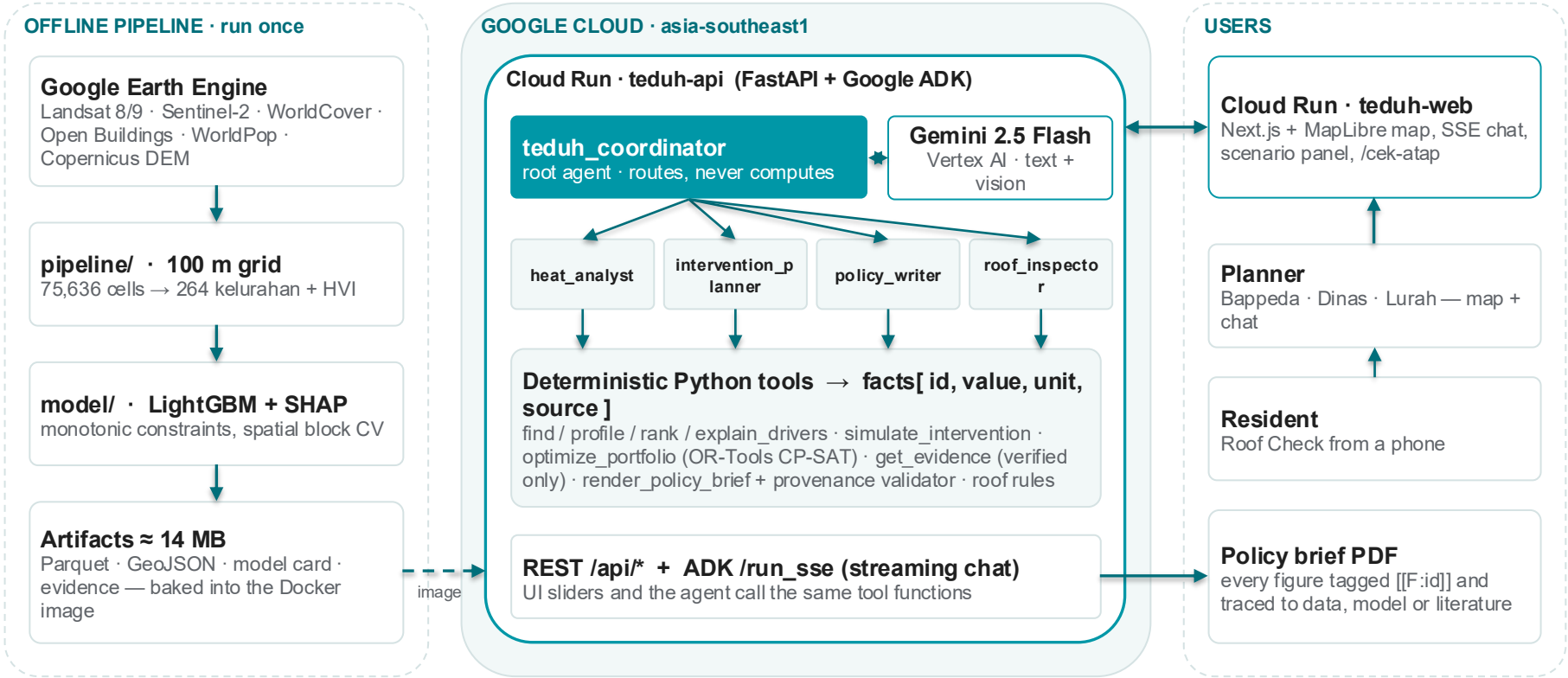
Process flow diagram or Use-case diagram



1. The planner enters an area, budget, or question.
2. The agent retrieves heat profiles, HVI, and key drivers.
3. The model estimates LST changes for intervention options.
4. The solver selects a package based on weighted benefits and equity constraints.
5. The validator checks number provenance; the brief is exported as a PDF.

Core principle: Gemini orchestrates and explains; every figure is calculated by deterministic Python tools, not freely generated by the language model

Architecture diagram of the proposed solution



Technologies to be used in the solution

AI & Agents

Gemini 2.5 Flash on Vertex AI • Google ADK • Gemini vision

ML & Explainability

LightGBM (monotonic) • SHAP • scikit-learn • spatial block cross-validation

Backend & Hosting

Python 3.12 • FastAPI • Cloud Run • Cloud Build • Secret Manager

Data & Geospatial

Google Earth Engine • Landsat 8/9 • Sentinel-2 • WorldCover • Open Buildings • WorldPop

Optimization

Google OR-Tools CP-SAT (budget + equity) • pymoo NSGA-II (roadmap)

Interface & Storage

Next.js • TypeScript • Tailwind • MapLibre • artifacts baked into the image

Evidence & Reporting

Verified evidence base • number-provenance validator • Markdown → PDF (WeasyPrint)

Model and region verified on Vertex AI: gemini-2.5-flash in asia-southeast1 (Singapore).

Estimated implementation cost (optional)

Running TEDUH for one pilot city · per month

teduh-api · Cloud Run, 2 vCPU / 2 GiB, 1 warm instance	≈ USD 26
teduh-web · Cloud Run, 1 vCPU / 0.5 GiB, 1 warm instance	≈ USD 10
Gemini 2.5 Flash · 1,000 planner questions (~20k input + 1k output tokens each)*	≈ USD 9
Earth Engine pipeline · offline, once per season; open datasets	USD 0 licence
Total, lower bound	**≈ USD 45**

Google Cloud list prices: Cloud Run idle min-instance USD 0.0000025 per vCPU-second and per GiB-second; Gemini 2.5 Flash USD 0.30 / 2.50 per 1M input / output tokens. Excludes active-request time, the Tier-2 regional rate, tax and egress. *Token volume is an assumption, not a measurement.

Unit costs the optimizer uses

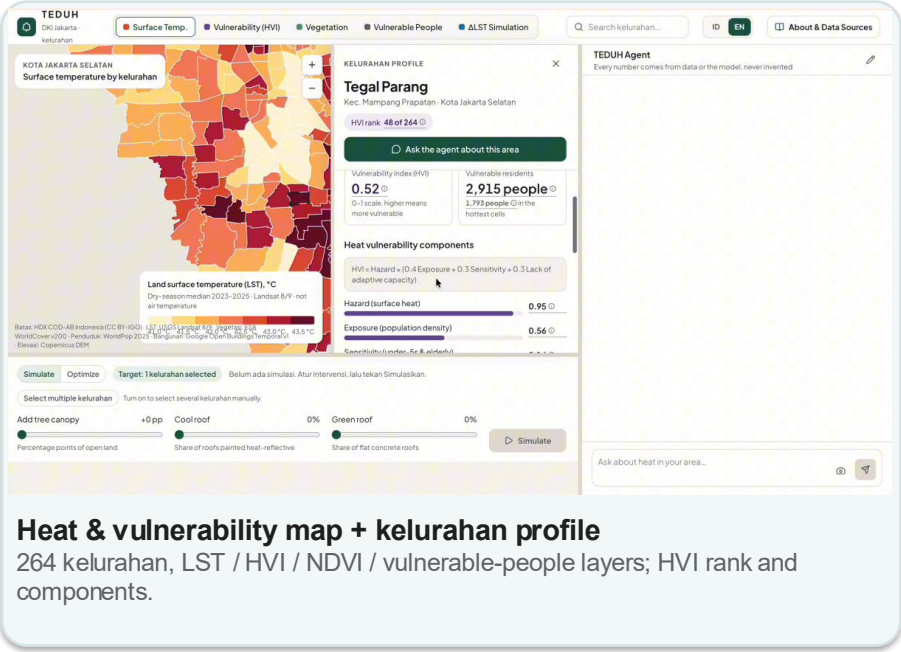
Shade tree, Tabebuya Ø > 20 cm Kepgub DKI 811/2025	Rp 2,702,703
Tree handling (rootball, proxy) Kepgub DKI 811/2025	Rp 1,950,000
Cool roof coating, per m² assumption · vendor quote pending	Rp 95,000
Green roof (extensive), per m² assumption · vendor quote pending	Rp 600,000

Live example · Central Jakarta, Rp 50 bn

Agent built a 5-kelurahan cool-roof portfolio for Rp 49.9 bn, 100% of it in the top HVI quintile, reaching ≈ 9,200 vulnerable residents in cooled cells (indicative LST).

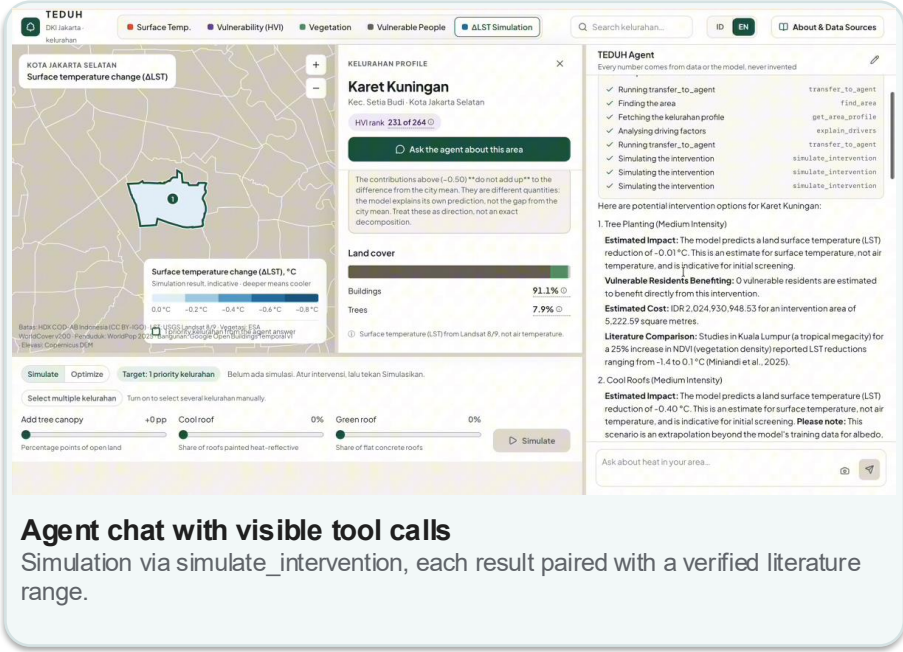
Snapshots of the prototype

Live on Cloud Run · captured from the deployed app (English UI)



Heat & vulnerability map + kelurahan profile

264 kelurahan, LST / HVI / NDVI / vulnerable-people layers; HVI rank and components.

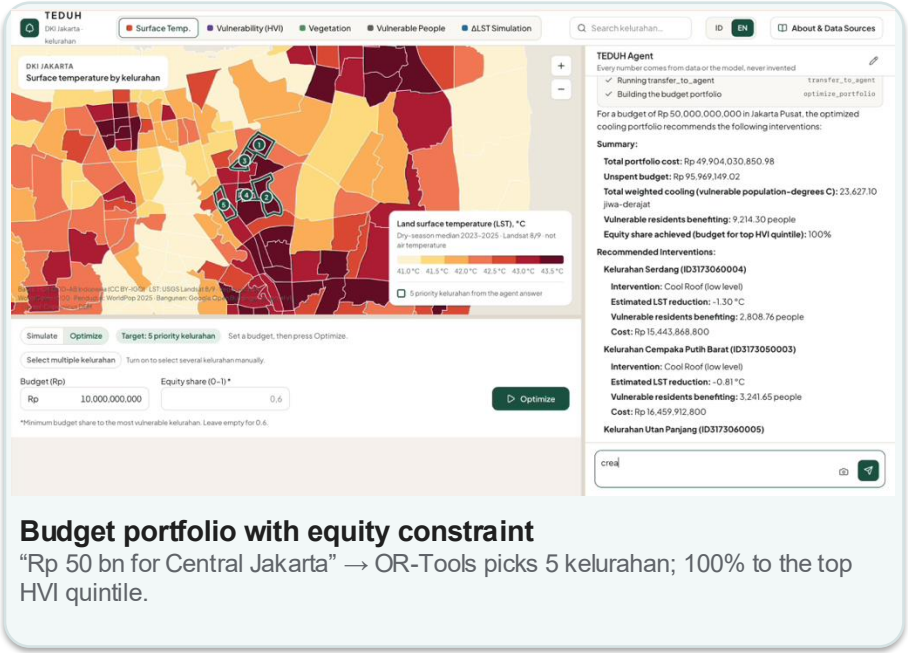


Agent chat with visible tool calls

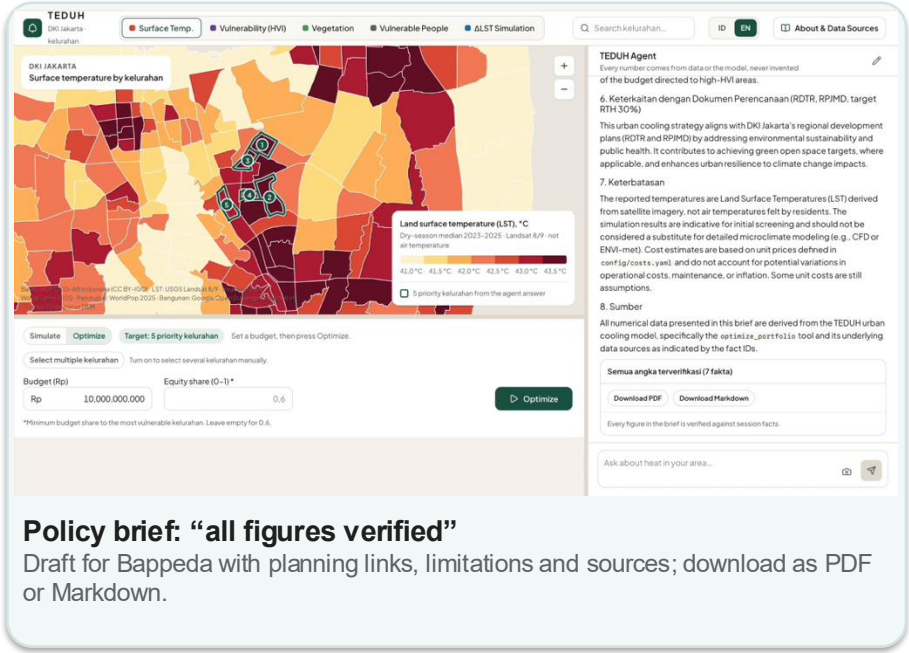
Simulation via simulate_intervention, each result paired with a verified literature range.

Snapshots of the prototype

From question to accountable policy document in one conversation

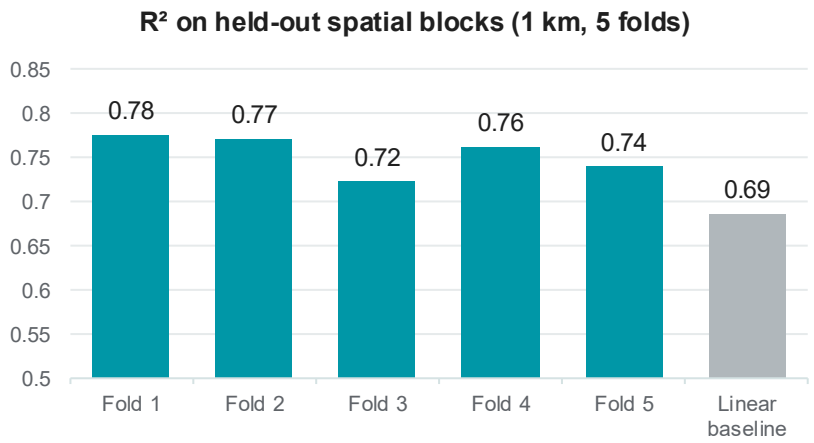


Budget portfolio with equity constraint
“Rp 50 bn for Central Jakarta” → OR-Tools picks 5 kelurahan; 100% to the top HVI quintile.



Policy brief: “all figures verified”
Draft for Bappeda with planning links, limitations and sources; download as PDF or Markdown.

Prototype Performance report/Benchmarking



0.755
mean R², spatial CV
folds 0.72 – 0.78

1.24 °C
RMSE
MAE 0.94 °C

+0.069
R² over linear baseline
0.686 → 0.755

75,636 cells × 264 kelurahan, dry season 2023–2025. Lat/lon deliberately excluded; random splits not reported because neighbouring 100 m cells leak.

0.54 % WorldPop vs Dukcapil population

0.14 °C BMKG vs GHCN air-temp cross-check

262 automated tests passing

100 % physics, provenance & budget/equity tests

Still below target: agent tool selection 65% (13/20, target 90%) · chat p50 9.4 s on a local run, 6.7–7.3 s warm on Cloud Run (target < 8 s) · Roof Check accuracy not yet measured (0 of 30 labelled photos).

Additional Details/Future Development (if any)

Next (P1)

- Pareto optimisation (NSGA-II): cost vs cooling vs equity
- Citizen Roof Check reports as an anonymous map layer
- Before / after split-map for scenarios
- Export the brief to Google Docs

Roadmap (P2)

- Citizen heat reports to correct satellite data
- Cool pavement from the OSM road network
- Seasonal heat alerts for posyandu cadres
- Air temperature at street level (PINN surrogate)

Adoption & scale

- Pilot with one city government (DKI Jakarta or Depok)
- Brief as an attachment to Musrenbang and RTH studies
- Global datasets only: a new city needs boundaries + period
- 500+ Indonesian cities and regencies; SDG 3, 11, 13

Gaps we close first

Vendor quotes for cool and green roofs (the largest remaining assumption) · verify the 6 remaining literature entries (4 of 10 verified today) · collect 30 labelled roof photos to report Roof Check accuracy · lift agent tool selection to ≥ 90% and chat p50 latency under 8 s.

Provide links to your:

- GitHub Public Repository : https://github.com/ahmadsyafiqkamil/teduh
- Demo Video Link (3 Minutes) : https://youtu.be/6eDuZJqVK7A
- Final Product Link : https://teduh-web-ibnno2v3ma-as.a.run.app/

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Thank you

